

Introduction to Cognitive Science

SITE - 5297

Introduction

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Outline

- Course Syllabus
- Cognitive and Neural Science - an Introduction

Learning Outcomes

- Upon successful completion of this course:
 - Understand **theory in cognitive psychology**, including perception, attention, learning, memory, reasoning, problem-solving, and decision-making
 - Understand **computational modeling of cognitive processes**
 - Understand the **biological bases of cognitive functions**
 - Critically evaluate **research and theory in cognitive and neural science**;



Topics to be Covered

Introduction to
Cognitive Sciences



Computational Model
of Mind



Philosophy of Mind



Vision and Color
Perception



Language Processing



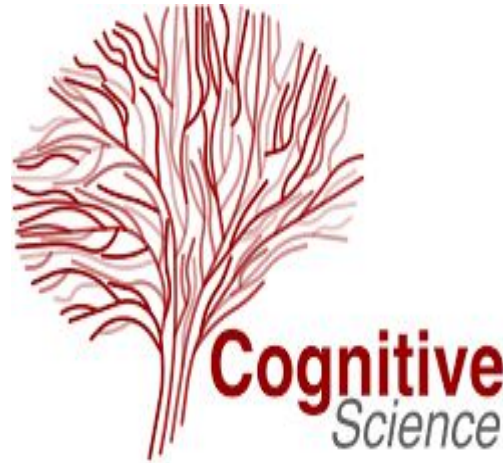
Machine Reasoning



Cognitive Development



Evolution of
Mind



Topics to be covered in Lab Sessions

- Introduction to selected Libraries
- Memory and Attention Lab – LSMT
- Cognitive Modeling and Machine Learning
- Natural Language Processing
- Computer Vision Lab

Administration

- **Teaching method:** Lecturing, classroom discussion, Lab exercises,
- **Course schedule:**
 - Lecture Session: Wednesday 1:30 – 3:30pm
 - Lab Session: TBA 1:30- 3:30 pm
- **Prerequisites:** Discrete Mathematics, Data structure and Algorithms

Books and References

- Reference:

- Gazzaniga, M. S. (Ed.). (2009). *The Cognitive Neurosciences* (4th ed.). MIT Press.
- Cambridge Handbook of Intelligence, R. Sternberg & S. Kauffman (editors), Cambridge university Press, 2011
- The Cognitive Sciences: An Interdisciplinary Approach by Carolyn P. Sobel and Paul Li. 2013.
- Kandel, E. R., Koester, J. D., Mack, S. H., & Siegelbaum, S. A. (Eds.). (2021). *Principles of Neural Science* (6th ed.). McGraw Hill.
- Sobel, Carolyn Panzer. *The Cognitive Sciences : An Interdisciplinary Approach*. 2nd ed. Los Angeles: SAGE, 2014. Print.
- Scientific papers and additional textbooks are also a plus for you

Administration

- **Course Policies**

- **Attendance:** Attendance is mandatory and punctuality is expected for all sessions. Missing more than 10% of lectures will result in a requirement to readmit for the course. Additionally, 100% attendance is required for all lab sessions.
- **Test/Quizzes:** Rarely reexamination schedules will be arranged for those who missed the exam by accidental or uncontrollable situation.
- **Cheating/Plagiarism:** No second Chance or excuses.
- **Generative AI Policy:** You may use Large Language Models (LLMs) for up to 30% of your assignment, but not beyond that limit.

Getting a Grade

- Quizzes 10%: Quiz I and II will be given without prior notice
- Mid-Exam 25%: Scheduled inline with the AAU Academic Calander
- Project Demonstration– 25% – group projects will be presented by ... due date May 15, 2026
- Final Exam – 40% –..... based on the academic calendar of the school



Introduction to Cognitive Science

Topics

- The mind
- What is Cognitive Science
- Representation

The Mind

- For centuries science has made great strides in our understanding of the external observable world.
- There were still many unanswered questions about something perhaps even more important to us.

“That something is the human mind”

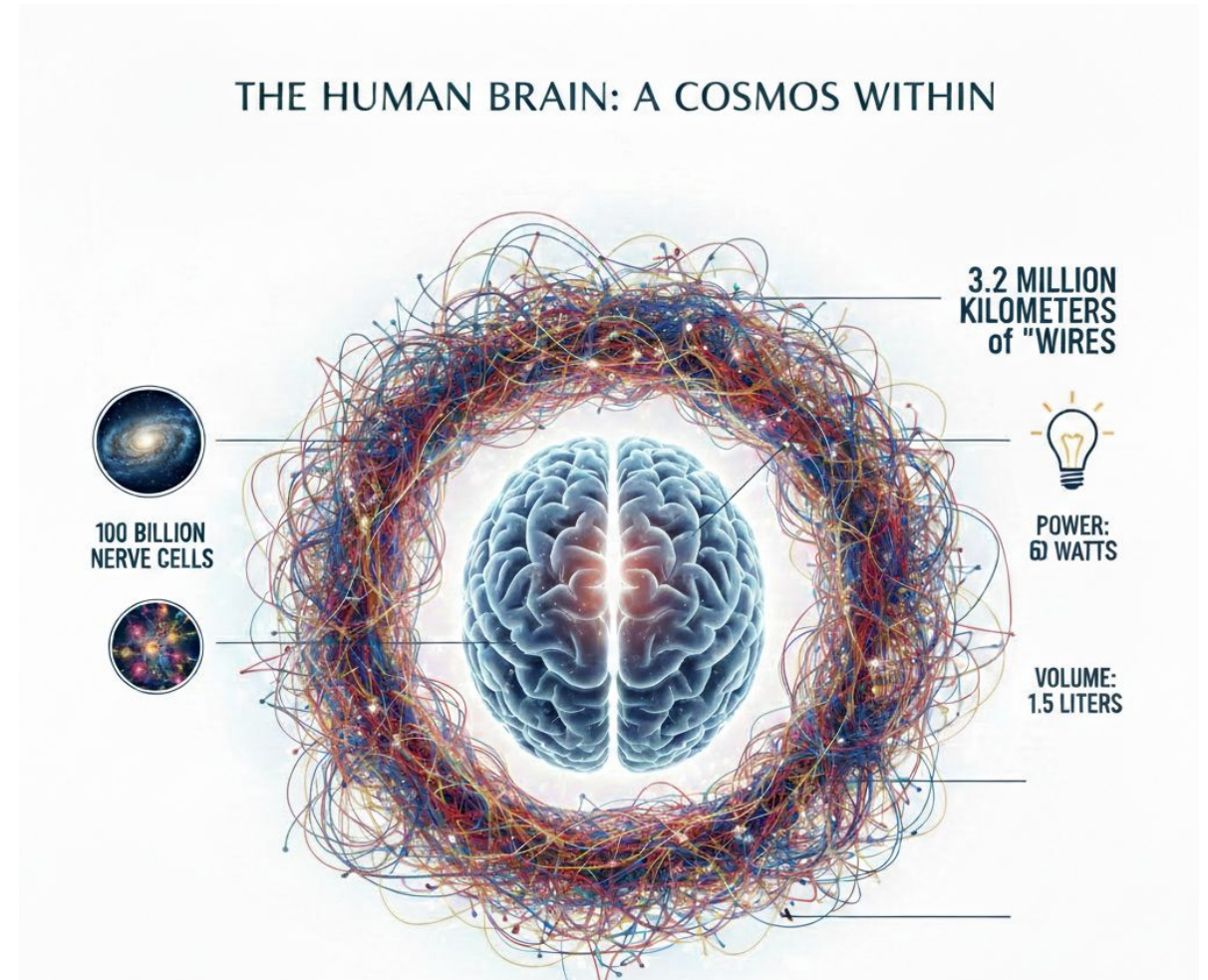
- What makes mind so difficult to study?

The Mind

- The mind is the most complex entity in the known universe.

“ Our brain contains roughly 100 billion nerve cells, 3.2 million kilometers of ‘wires’, a million-billion connections, all packed into a volume of 1.5 litres, but weighing only 1.5 kg and consuming a mere 10 watts.”

- Each neuron can make about 1000 connections— used 100 trillion data points (100 terabytes of information)--- web like connections
 - Longer than the total number of atomic particles that make up the known universe
 - Many agree on 10 terabytes of brain storage capacity which is equivalent to 470 million books



Brain vs Mind?

The Mind

Brain	Mind
Brain is the central organ of the human nervous system	Mind is a faculty that manifests itself in mental phenomena such as perception, thinking, sensation, reasoning, memory, etc.
It is made up of blood vessels and nerve cells	It is not made up of any cells and is hypothetical.
It has a definite shape and structure	It does not have a defined share and structure.
Brain coordinates, movements, feelings, and different functions of the body	It refers to a person's conscience, understanding and thought process.
You can touch the brain	You cannot touch the mind.

Questions

- What is cognitive science?

Cognitive Science: Interdisciplinary Approach

- Cognition – from Latin base *cognitio* – “know together”
- The collection of mental processes and activities used in perceiving, learning, remembering, thinking, and understanding and the act of using those processes
- Cognitive process: Learning and Memory, motivation, Attention, emotion, Thinking and Reasoning (Planning, Decision Making, Problem Solving ...), Language,, Vision-Perception, Social Cognition, Dreaming and Consciousness

Cognitive Science: Interdisciplinary Approach

"Cognitive science is the interdisciplinary study of mind and intelligence, embracing philosophy, psychology, artificial intelligence, neuroscience, linguistics, and anthropology."

- **Source:** Stanford Encyclopedia of Philosophy
- Possible definitions:
 - "Cognitive science can be roughly summed up as the scientific interdisciplinary study of the mind".
 - "Scientific study of the mind, the brain, and intelligent behaviour, whether in humans, animals, machines or other abstract"

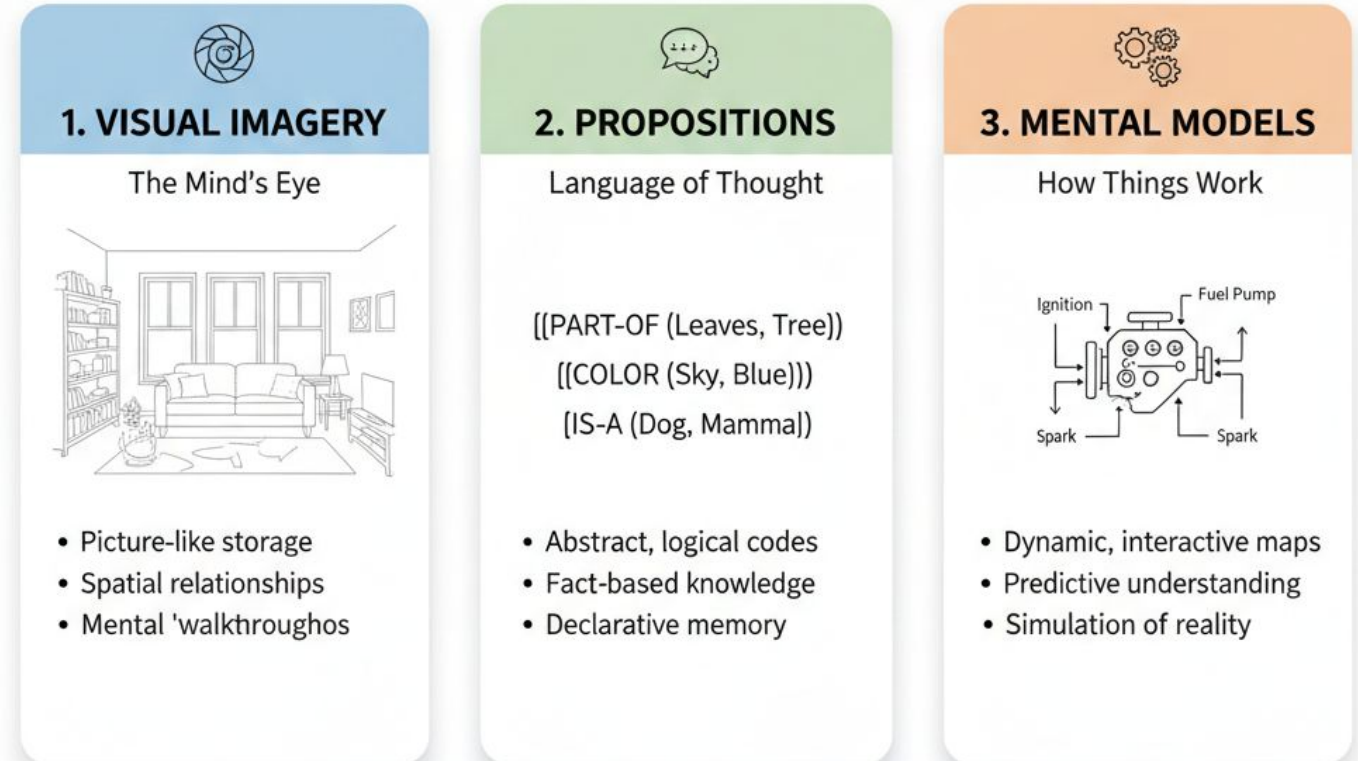
Cognitive Science

- In order to really understand what cognitive science, its theoretical perspective centers on the idea of **computation**, called information processing.
- Cognitive scientists view the mind as an information processor.
- Cognitive science is often credited with being influenced by the rise of the computer-- at a very high level of abstraction
- A mind, accordingly, must incorporate some form of mental representation and processes that act on and manipulate that information.

Representation

- **Mental representations** are crucial for memory recall; they allow us to retrieve past experiences and knowledge when faced with new situations.
- **Dynamic** and can change over time as new experiences or information are integrated into our understanding.
- Different types of mental representations include **imagery (visual)**, **linguistic (verbal)**, and **conceptual (abstract)**, each serving distinct functions in cognition.
- The ability to create and manipulate mental representations is essential for higher-order thinking tasks such as planning and problem-solving.

How the Mind Stores Reality



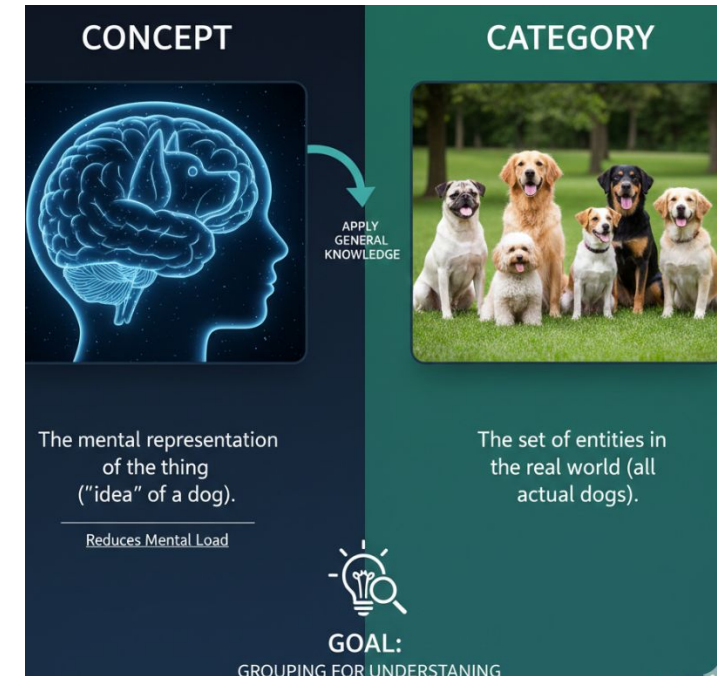
Source: Cognitive Science Primer - 2026

Representation

- Aspects of any representation
 - Representation must be realized
 - Representation must have content--- The thing or things in the external world that a representation stands for are called referents.
 - A representation must also be “grounded.”
 - A representation must be interpretable by some interpreter
- Representation:- digital, analog, dual-coding hypothesis, and propositional representation

Representation: Concepts and Categories

- Cognitive psychologists have paid a good deal of attention to the subject of concepts and categories
- Concepts and categories are the "building blocks" of thought.
- **Concept:** The mental representation of a thing (the idea of a dog).
- **Category:** The set of entities or objects in the real world that the concept refers to (all the actual dogs that exist).
- **The Goal:** By grouping objects so we can apply general knowledge to new instances.



Concepts and Categories

- If we hope to understand how our minds work, we need to what factors in our experience contribute to forming our categories and what part of our cognitive equipment is involved.

Mental Representation

- Our concepts and categories are stored somehow in our brain
- We construct categories from instances and these instances result in concepts.
- Often concepts are related:
 - In 350 BCE Aristotle described, people pass swiftly in thought from one point to an other, from milk to white from one starting point to several directions

Mental Representation

- **Associations** are due to the shared elements in the associated thoughts.
- A **proposition** that surfaces at a **conscious** level may be draw out of memory by the repetition of arguments.
- Without memory there is **no propositional representation**.... depends crucially on our ability to store them over long periods.
- The associative process works is that meaning and links are stored in a **proportional, or semantic network**
- **Semantic network** are represented by **nodes**.... the association between nodes are represented by lines, links.... links are **weighted** to show how essential to the meanings of various concepts

Propositional Network

- When we think of something-anything-it is likely to be couched in what has been termed a **proposition**.
- A **proposition** is a unit of meaning that can take a **truth value**, a unit of meaning that can be either true or false.
 - Expressed in language.
 - Is important in language translation
- The concept standing in relation to each other are called arguments of the relation.
- A **Propositional Network** is a mental model that represents knowledge as a complex web of interconnected nodes, where each connection signifies a **specific relationship between concepts**

Computation

- Representations are only the first key component of the cognitive science view of mental processes.
- Representations by themselves are of little use unless something can be done with them.

“Having the concept of money doesn’t do much for us unless we know how to calculate a tip or can give back the correct amount of change to someone.”

- In CS, the mind performs computations on representations. It is therefore important to understand how and why these mental mechanisms operate.
- What sorts of mental operations does the mind perform?

Computation

- **The Tri-Level Hypothesis**

- Mental or artificial information-processing events can be evaluated on at least three different levels
- The highest or most abstract level of analysis is the **computational level**, two tasks
 - Describing the problem in a precise way
 - Reason for the process
- Explanatory perspective used in the evolutionary approach
- **Algorithmic level** a formal procedure or system that acts on informational representations.
- The most specific and concrete type of description is formulated at the **implementational level**.

Computation: Mathematical concepts used to study mind representation

- Linear Algebra

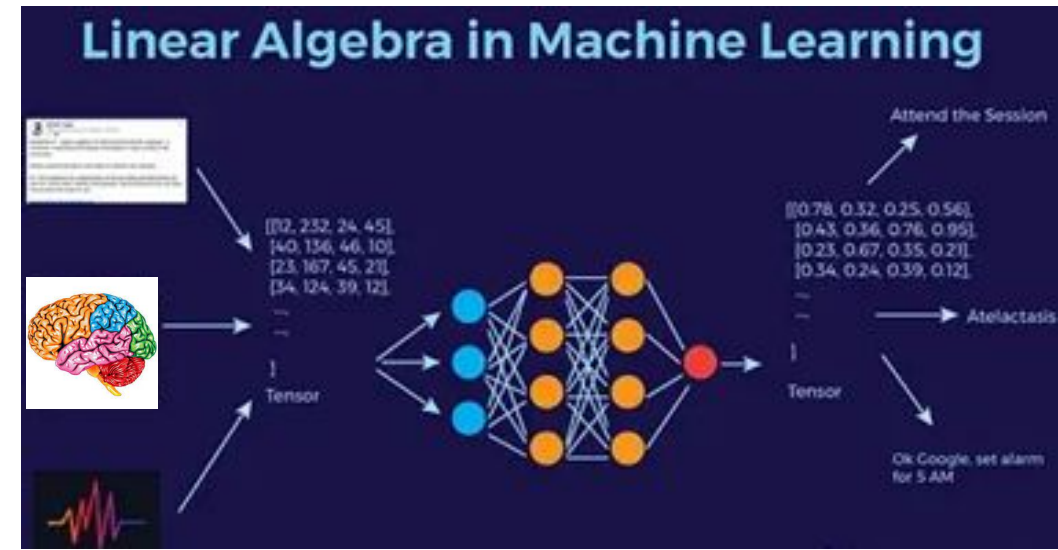
- Vectors and Matrices:

- Every piece of information is converted into a vector.
 - The connections between neurons (represented by the orange nodes in the image) act as matrices.
 - Neural networks in the brain can be modeled as matrices that transform input vectors into output vectors.

- Eigenvalues and Eigenvectors: Help analyze the stability and dynamics of neural systems.

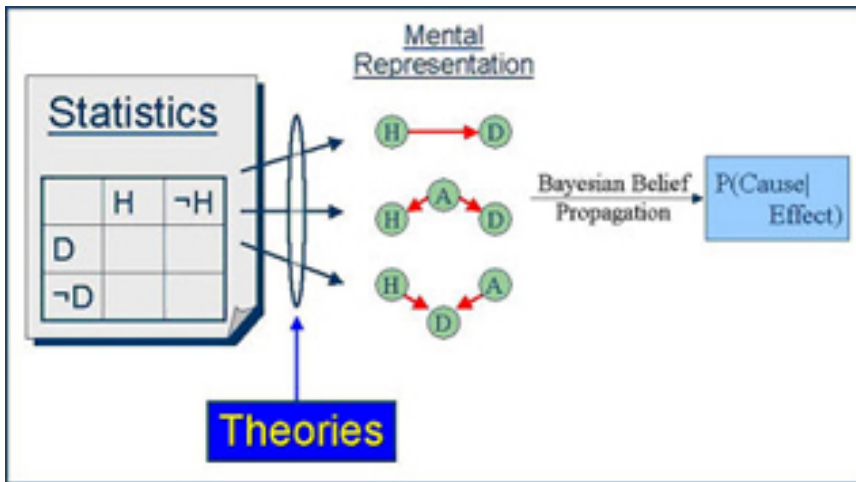
- Eigenvectors: Represent the "preferred directions" of a neural system
 - Eigenvalues: Determine how much weight or "strength" these patterns have.

- Principal Component Analysis (PCA): The brain receives millions of sensory inputs every second. PCA is the mathematical tool used to filter out the "noise" and focus on the signal.



Computation: Mathematical concepts used to study mind representation

- **Probability and Statistics:** while linear algebra provides the anatomy of a representation (the vectors and matrices), Probability and Statistics provide the logic of belief and decision-making.
- **Bayesian Inference:**
 - The mind doesn't just see facts; it maintains a set of hypotheses (H) about the world. When new data (D) arrives, it uses Bayesian Inference to update the probability that a specific hypothesis is true.
 - Models how the brain updates beliefs or representations based on new evidence (e.g., predictive coding theories).
 - The diagram shows $P(\text{Cause} | \text{Effect})$. This represents how we infer the hidden cause of a situation based on visible symptoms.

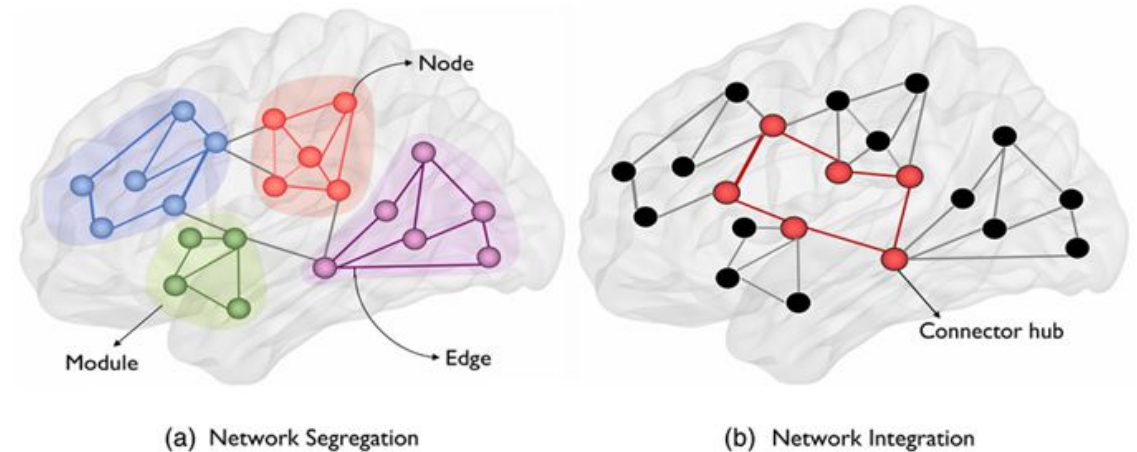


Computation: Mathematical concepts used to study mind representation

- Probability and Statistics
- CS suggests that Mental representations are rarely black and white. Instead, they are modeled as Probability Distributions.
 - Handling Ambiguity: When you see a blurry shape in the dark, your brain doesn't just "guess" one object. It calculates a distribution of possibilities (e.g., 70% it's a coat rack, 30% it's a person).
 - Optimal Inference: This aligns with the Bayesian Brain Hypothesis, which proposes that the brain is a statistical organ that minimizes surprise by constantly comparing sensory input against its internal distributions.

Computation: Mathematical concepts used to study mind representation

- Graph Theory
 - **Networks and Connectivity**: Represent the brain as a graph, with nodes (neurons or brain regions) and edges (connections). This helps study the structure and function of neural networks.
 - **Centrality Measures**: Identify key nodes or regions in the brain that play critical roles in information processing.
- Topology
 - **Manifolds**: CS uses **Manifold Learning** to explain how **high-dimensional sensory data** is compressed into a **lower-dimensional mental map**. This allows us to navigate physical spaces or abstract concepts efficiently.
 - **Persistent Homology**: The brain is a high-dimensional system-- **persistent homology** allows cognitive scientists to find **holes or structures** in neural data that persist across different scales of observation.



Computation: Mathematical concepts used to study mind representation

- **Game Theory**:- in social cognitive science, the brain is modeled as a player in a game,
 - **Strategic Decision-Making**: Models how the mind represents and predicts the behavior of others in social interactions.
 - **Nash Equilibrium**: Describes stable states in decision-making processes.
 - Used to study how social norms and cooperation become "hard-wired" into our behavioral representations.
- **Signal Processing**
 - **Fourier Transforms**: Analyze neural signals in the frequency domain.
 - **Filtering and Noise Reduction**: Help understand how the brain processes noisy sensory inputs.

Mental Representation

- Understanding mental representations through **mathematics** can be applied to various fields including:
 - **Cognitive development**: How children acquire and develop mental representations of concepts.
 - **Decision making**: How people represent options and make choices based on their mental models.
 - **Language processing**: How words and sentences are represented in the mind.
 - **Memory and retrieval**: How memories are encoded and retrieved based on mental representations.

Types of Categorization: The Exemplar Approach

- As we grow up, we are presented with bits and pieces of the world around us, from which we must construct the categories that allow us to function in the world.
- Our experience allows us to do categorization.... called exemplars.
 - Example: If you see a new animal, your brain quickly compares it to your stored exemplars.
 - Cognitive Mechanism: If the new animal is "mathematically close" to your stored exemplars, your brain triggers the category: "That's a bird."
- **Challenge:** Categorization among entities assigned to a particular category might be different from each other
 - Two animals share almost no visual traits. If you only had one prototype of a bird, you might fail to recognize one of them.

What solution do you propose?

Types of Categorization: The Feature Approach

- Is the opposite of the Exemplar Theory -- instead of looking at specific memories, it looks for a checklist of traits.
- Requires only a few clear-cut identifying features to characterize it.
 - Features: 4 sides + 4 right angles + all sides equal.
 - Result: If a shape meets those three features, it is a square. No exceptions.
- Identifying the necessary and sufficient features for any category one might choose is not always as easy as it is.

Types of Categorization: The Prototype Approach

- A third model of categorization **resets on the notion of norm**, a model based on **prototype**.
- The **prototype approach** is the brain's way of creating a **mental average....** is a version/representation of an **entity formed of average values for the features characterizing the entity**.
- Instead of remembering every bird you've seen, your brain blends them all together into one generic bird.
 - Given: It has a medium-sized beak, medium wingspan, and can fly.
 - The Example: If I ask you to Think of a bird, you likely think of something like a Robin or a Sparrow (the prototype), rather than a Penguin or an Ostrich (the outliers).
- **Categorizing** by the prototype method is categorizing **heuristically**, not complete search, search only for similarity to prototypes.

The Interdisciplinary Perspective

Interesting metaphor for cognitive science

- There is an old fable about five blind men who stumble upon an elephant

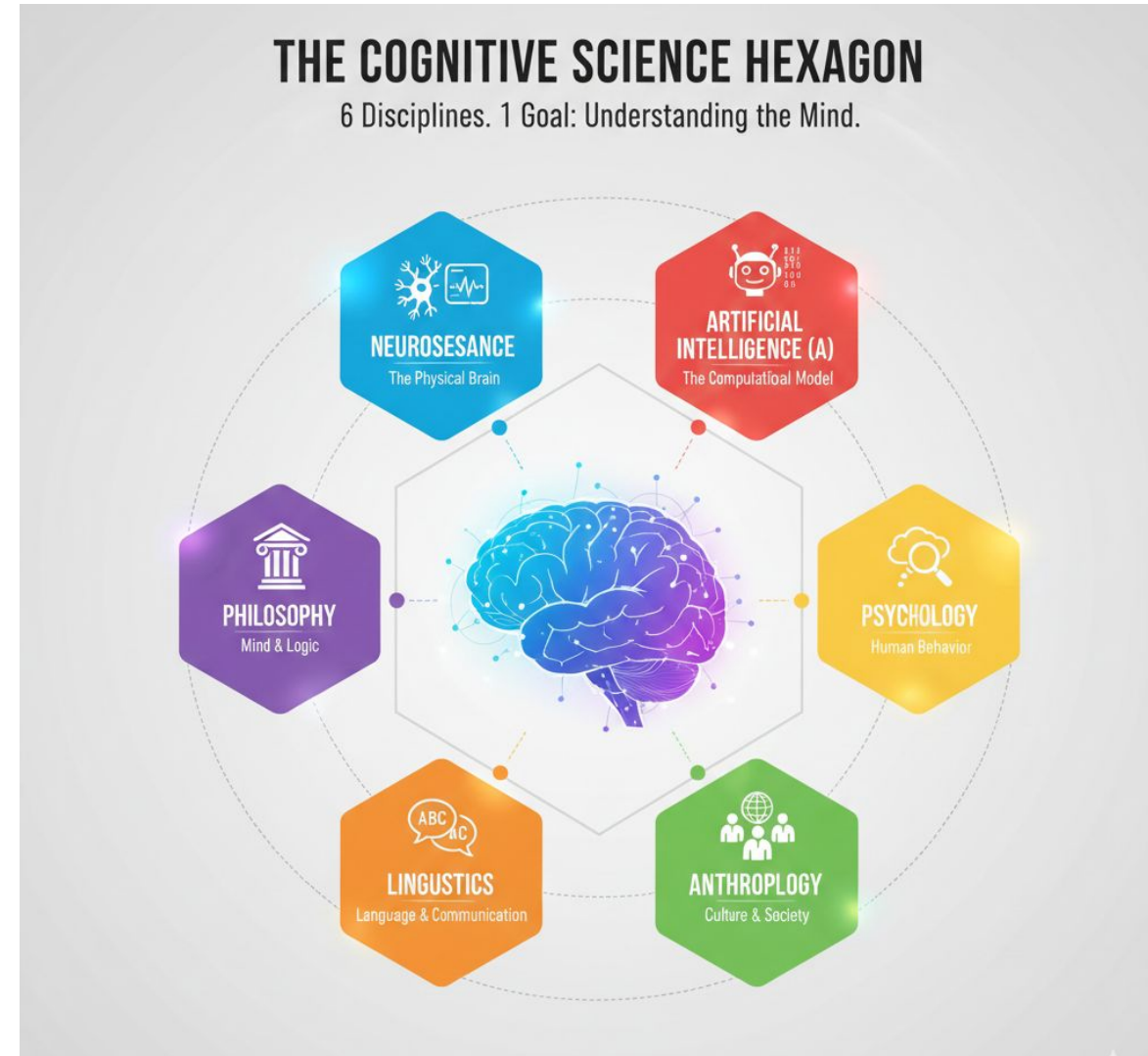


- All five men are wrong in their conclusions because each has only examined one aspect of the elephant (a tusk, a leg, a tail)
- If the five men had gotten together and shared their findings, they may have easily pieced together what kind of creature it was.

The key to figuring out something as mysterious and complex as mind is communication and cooperation among disciplines.... Shows how complex systems—like the human mind—cannot be understood through a single lens.

The Interdisciplinary Perspective

- **Neuroscience:** It studies the physical-hardware... which part of the brain lights up when we feel happy?
- **Artificial Intelligence:** AI experts build computer models to simulate how humans process data.
- **Linguistics (The Code):** Language is the primary way we express our thoughts. This field looks at the rules of communication. It asks: *How does the brain turn a thought into a spoken sentence?*
- **Anthropology (The Environment):** This studies how culture and society shape our thinking. It asks: *Does a person living in a digital society think differently than someone in a hunter-gatherer tribe?*



Approaches in Cognitive Science: Philosophical Approach

- Oldest of all the disciplines in cognitive science, dated back to the ancient Greeks.
- **Question** on virtually any subject, ranging from the nature of existence to the acquisition of knowledge, to politics, ethics, and beauty.
- The primary method of philosophical inquiry is reasoning, **both deductive and inductive**.
- **Philosophy (The Concept)**: Philosophy deals with the "big questions" that math can't always solve, like the nature of consciousness or the definition of "thought." It asks: *Is the mind just a machine, or is it something more?*

Approaches in Cognitive Science: Psychological Approach

- Psychology arose in the late 19th century and was the first discipline in which the scientific method was applied exclusively to the study of mental phenomena.
- Psychologists apply the scientific method to both mind and behavior.
- Psychoanalytic psychology conceives of mind as a collection of competing entities, while behaviorism sees it as a device that maps stimuli onto behaviors.
- **Psychology (The Behavior):** This focuses on how humans actually act and react. It asks: Why do we remember some things perfectly but forget others?

Approaches in Cognitive Science: The Cognitive Approach

- Starting in the 1960s a new form of psychology arrived on the scene. Known as cognitive psychology
- Cognitive psychologists placed renewed emphasis on the study of internal mental operations.
- Adopted the computer as a metaphor for mind, and described mental functioning in terms of representation and computation
- Was also better able to explain phenomena such as language acquisition, for which behaviorists did not have good accounts.
- Inherent in the cognitive approach is the idea of modularity
- Cognitive psychologists have studied a wide variety of mental processes, including pattern recognition, attention, memory, imagery, and problem-solving.

Approaches in Cognitive Science: The Neuroscience Approach

- The general study of the brain and endocrine system is called neuroscience.
- The attempt to explain cognitive processes in terms of underlying brain mechanisms is known as cognitive neuroscience.
- Neuroscience, first and foremost, provides a description of mental events at the implementational level.
- It attempts to describe the biological “hardware” upon which mental “software” supposedly runs.
- There are many levels of scale when it comes to describing the brain, and it is not always clear which level provides the best explanation for any given cognitive process.

Approaches in Cognitive Science: The Neuroscience Approach

- Neuroscientists, investigate at each of these levels.
- Neuroscientists employ a wide variety of machines to measure the performance of the brain at work
 - Machines: positron emission tomography (**PET**) scanners, computerized axial tomography (**CAT**) scanners, and magnetic resonance imaging (**MRI**) machines.
- Neuroscientists use other techniques as well-- use single- and multiple-cell recording techniques.

Neuroscience: It studies the physical-hardware... which part of the brain lights up when we feel happy?

Approaches in Cognitive Science: The Network Approach

- Partially derived from neuroscience
- Mind is seen as a collection of individual **computing units**-- connected to one another and mutually influence one other's activity via the connections.
- The connectivity of the units can give rise to representational and computational complexity

Approaches in Cognitive Science: The Evolutionary Approach

- Natural selection holds that adaptive features enable the animals that possess them to survive and pass these features on to future generations.
- The field of evolutionary psychology applies selection theory to account for human mental processes.
- It attempts to elucidate the selection forces that acted on our ancestors and how those forces gave rise to the cognitive structures we now possess.
- Evolutionary theories have been proposed to account for experimental results: from categorization to memory, to logical and probabilistic reasoning, language, and cognitive differences between the sexes.
- A variant on this theme is evolutionary computing, in which the rules of evolution are applied to create successful computer algorithms.

Evolutionary psychologists also adopt a modular approach to mind.

Approaches in Cognitive Science: The Linguistic Approach

- Linguistics is an area that focuses exclusively on the domain of language.
- Part of the difficulty in studying language is the fact that language itself is so complex.
- **Linguistic investigation:** primate language use, language acquisition, deficits in language acquisition caused by early sensory deprivation or brain damage, the relationship between language and thought, and the development of speech recognition systems.

Approaches in Cognitive Science: The Artificial Intelligence Approach

- In the past few decades that computer scientists have seriously attempted to build devices that mimic complex thought processes--- artificial intelligence (AI)
- AI secondarily gives us insights into the function of human mental operations.

“An even more exciting outcome of AI research is that someday we may be able to create an artificial person who will possess all or many of the features that we consider uniquely human, such as consciousness, the ability to make decisions, and so on.”

Approaches in Cognitive Science: The Robotics Approach

- The robotics approach has much to contribute to cognitive science and to theories of mind.
- Robots, like people and animals, must demonstrate successful goal-oriented behaviors under complex, changing, and uncertain environmental conditions.
- Robotics therefore helps us to think about the kinds of minds that underlie and produce such behaviors.

Cognitive Science vs AI

- "Cognitive science focuses on human minds with artificial minds as an example".
- "AI focuses on artificial minds with human minds as an example".

Source Max Louwerse: How Cognitive Science and Artificial Intelligence are Intertwined. June 20, 2022

Thank You!